

Table of Contents

Highlights

You have learned how to use the distributive property to multiply each term inside parentheses by the monomial outside parentheses.

$$4x(x^2 - 6)$$

First Outer
Inner Last

$(x + 3)(x + 8)$

Multiply the terms in this order, and then combine like terms to simplify the expression.

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Step 1

First: $x \bullet x = x^2$

Outer: $x \cdot 8 = 8x$

Inner: $3 \cdot x = 3x$

Last: $3 \cdot 8 = 24$

Write the expression.

$$x^2 + 8x + 3x + 24$$

Combine like terms.

$$x^2 + 11x + 24$$

$$(x + 3)(x + 8) = \mathbf{x^2 + 11x + 24}$$

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Step 1

First: $2y \cdot y = 2y^2$

Outer: $2y \cdot -6 = -12y$

Inner: $-1 \cdot y = -y$

Last: $-1 \cdot -6 = 6$

Write the expression.

$$2y^2 - 12y - y + 6$$

Combine like terms.

$$2y^2 - 13y + 6$$

$$(2y - 1)(y - 6) = 2y^2 - 13y + 6$$

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Category

Lesson 1: Polynomial Operations

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